

MUONIO ALAMUONIO, Finland

WMO: 028230

Lat: 67.9583N

Lon: 23.6847E

Elev: 829

StdP: 14.26

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.3	-22.7	-33.5	0.9	-28.2	-27.3	1.2	-22.6	18.7	23.7	16.5	23.5	1.5	330	0.711

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	75.4	61.1	71.4	59.2	67.7	57.6	63.1	71.2	60.9	68.8	58.8	65.5	6.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.1	79.9	65.6	57.7	73.2	63.0	55.8	68.3	61.5	29.0	71.3	27.4	68.9	26.0	65.6	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	13.8	12.0	DB	-33.0	79.2	7.3	3.2	-38.2	81.5	-42.5	83.3	-46.7	85.1	-52.0	87.4
			WB	-33.2	65.5	7.3	2.5	-38.4	67.3	-42.7	68.7	-46.8	70.1	-52.1	71.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	30.9	7.0	8.2	17.6	29.3	40.8	51.4	57.7	53.1	43.7	30.9	18.1	11.2
	DBStd	20.36	14.56	14.44	11.06	8.10	7.69	6.90	5.91	6.17	6.65	8.63	12.21	14.54
	HDD50	7502	1332	1171	1005	621	304	63	6	38	207	593	958	1204
	HDD65	12471	1797	1591	1470	1071	751	410	237	371	638	1058	1408	1669
	CDD50	523	0	0	0	0	18	106	246	134	19	0	0	0
	CDD65	16	0	0	0	0	0	3	11	2	0	0	0	0
	CDH74	157	0	0	0	0	7	28	102	20	0	0	0	0
	CDH80	12	0	0	0	0	0	1	10	1	0	0	0	0
Wind	WSAvg	5.0	3.9	4.6	5.6	5.2	6.1	6.1	5.3	5.0	5.0	4.8	4.0	4.1
Precipitation	PrecAvg	22.2	1.4	1.2	1.1	1.2	1.6	2.5	3.0	2.9	2.0	1.9	1.8	1.5
	PrecMax	27.3	3.0	2.4	2.6	2.9	3.4	4.6	5.0	5.0	4.1	4.1	3.3	3.0
	PrecMin	16.2	0.2	0.2	0.3	0.3	0.3	0.4	1.0	1.3	0.4	0.4	0.6	0.4
	PrecStd	3.2	0.6	0.6	0.6	0.6	0.8	1.1	1.1	1.1	1.0	0.9	0.7	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.5	38.0	42.4	54.5	73.2	78.0	81.0	76.6	63.2	50.2	41.0	37.7
		MCWB	33.2	34.7	36.5	44.5	57.3	61.8	63.3	62.9	54.0	47.3	40.1	34.6
	2%	DB	31.6	32.9	38.3	48.2	65.6	72.2	76.9	71.8	59.4	47.7	36.4	33.6
		MCWB	30.8	31.2	33.5	40.6	52.7	57.9	62.0	60.6	52.8	45.4	35.5	32.6
	5%	DB	28.8	30.4	34.9	44.4	58.3	67.8	72.9	67.6	56.7	44.9	33.9	31.9
		MCWB	28.1	29.5	31.4	37.9	48.3	56.1	60.4	58.5	51.8	43.1	33.4	31.3
	10%	DB	26.0	27.7	32.1	41.0	53.5	63.7	68.8	63.7	54.1	42.0	32.3	30.0
		MCWB	25.4	26.7	29.7	35.8	45.2	53.5	58.8	56.6	50.6	40.4	32.0	29.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.3	35.1	36.7	45.1	57.8	63.7	66.4	65.3	57.2	49.0	40.6	35.1
		MCDB	34.5	37.4	41.7	53.8	71.0	73.3	74.9	72.6	60.8	49.4	40.7	36.8
	2%	WB	30.9	31.6	34.0	40.8	53.1	59.7	63.9	61.9	54.6	45.8	35.7	32.8
		MCDB	31.5	32.7	37.4	47.1	63.7	69.0	72.4	69.0	57.3	46.8	36.2	33.5
	5%	WB	28.3	29.5	31.9	38.4	49.3	56.9	61.9	59.5	52.9	43.2	33.5	31.5
		MCDB	28.8	30.5	34.3	43.6	56.8	65.3	69.8	65.2	55.6	44.7	33.9	31.9
	10%	WB	25.3	26.7	29.9	36.3	46.3	54.7	60.0	57.6	51.0	40.5	32.1	29.3
		MCDB	25.9	27.7	32.1	40.3	52.3	62.4	67.4	62.6	53.7	42.0	32.4	29.9
Mean Daily Temperature Range	5% DB	MDBR	14.8	14.7	18.5	16.8	14.9	14.2	15.1	15.4	13.1	9.9	11.3	13.7
		MCDBR	12.8	11.8	16.2	19.2	23.8	20.9	22.3	21.4	16.6	10.6	8.1	9.7
		MCWBR	12.7	10.6	12.5	12.6	13.1	10.4	10.3	11.0	10.6	8.7	7.7	9.4
	5% WB	MCDBR	12.5	11.8	13.7	17.2	21.0	18.0	18.4	17.6	13.3	9.3	7.5	9.6
		MCWBR	12.5	10.7	11.1	11.4	12.4	9.6	9.7	10.2	9.1	8.2	7.3	9.4
Clear-Sky Solar Irradiance	taub	0.188	0.248	0.284	0.314	0.333	0.337	0.346	0.331	0.300	0.267	0.351	N/A	
	taud	1.919	2.089	2.181	2.210	2.413	2.483	2.472	2.497	2.557	2.324	2.760	N/A	
	Ebn at Noon	35	174	238	264	273	275	268	260	242	173	50	0	
	Edh at Noon	3	15	25	33	30	29	29	25	18	12	4	0	
All-Sky Solar Radiation	RadAvg	12	161	641	1260	1555	1584	1485	1071	573	191	30	0	
	RadStd	1	18	36	59	152	110	146	78	66	31	4	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.15	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page